Supplementary Material 5

Relevant outcomes of included studies

First author, Year	Study design	Control group (type, *n* and % female, age(SD) years)	Sleep outcome	Carer burden measure	Sleep Results Mean (SD)	Carer Burden Results Mean (SD)
Bartolomei, 2018 [30]	Cross-sectional	-	MOSS-SS Index	CBI	MOSS-SS: 20.1 (18.1)	CBI: 9.0 (13.4)
Cifu, 2006 [35]	Cross-sectional	-	Self-report sleep duration (hours/per night)	ZBI CDS	Sleep duration: 6.4 (1.5)	ZBI: Not reported CDS: Not reported
Cupidi, 2012 [47]	Case-control	Healthy controls, *n* = 150, F = 35%, 66.7 (5.1)	PSQI	CBI	**Carer**; Global PSQI: 6.3 (3.9) Disturbance: 1.3 (0.5) Quality: 1.3 (0.9) Duration: 1.1 (1.0) Latency: 1.1 (0.9) Daytime dysfunction: 0.6 (0.7) Sleep efficiency: 0.8 (1.1) **Control;** Global PSQI: 4.3 (3.1) Disturbance: 1.0 (0.5) Quality: 0.9 (0.8) Duration: 0.7 (0.7) Latency: 0.7 (0.7) Daytime dysfunction: 0.8 (0.9) Sleep efficiency: 0.4 (0.6)	CBI: 13.4 (13.4)
Hand, 2021 [36]	Cross-sectional	-	SCOPA-Sleep	CBI	Not reported	CBI: 62.9% ≥ 24, 32.7% ≥ 36
Happe, 2002 [37]	Cross-sectional	-	Self-report poor sleep in last week	CBI	Self-report poor sleep: 27% of carers	Not reported

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Hermanowicz, 2019 [38]	Cross-sectional	-	PMDAlliance survey	-	Excessive tiredness carer: 84% of carers	-
Hicks, 2019 [39]	Cross-sectional	-	PSQI	-	Global PSQI: 4.9 (2.6)	-
Lee, 2020 [40]	Cross-sectional	-	MINI - Insomnia	-	Insomnia: 7.4% of carers	-
Liu, 2018 [32]	Cross-sectional	i) FTLD carers, *n* = 131, F = 37%, 58.3 (13.5) ii) AZ carers, *n* =325, F = 46%, 60.6 (12.7)	PSQI	ZBI	**PD Carers;** Global PSQI: 5.2 (3.0) **FTLD Carers;** Global PSQI: 4.4 (3.7) **AZ Carers;** Global PSQI: 1.01 (1.4)	**PD Carers;** ZBI: 22.6 (16.5) **FTLD Carers;** ZBI: 23.6 (15.9) **AZ Carers;** ZBI: 12.3 (9.7)
Lökk, 2008 [41]	Cross-sectional	-	Sleep disturbance is last four weeks (yes/no)	-	Sleep disturbance: 46%	-
Olsson, 2016 [29]	Case-control	Healthy controls, *n* = 79, F = 59%, 71.9 (9.0)	NHP - Sleep	-	**Carer Sleep;** md(q1-q3; min-max) 20 (0-60; 0-100) **Control Sleep;** 20 (0-40; 0-100)	-
Ozdilek, 2012 [42]	Cross-sectional	-	WHOQOL-BREF - Sleep	ZBI	Not reported	ZBI: 27.6 (15.1)
Pal, 2004 [43]	Cross-sectional	-	PSQI GSQ	-	PSQI; Global Score: 5.5 (3.8) Disturbance: 1.1 (0.5) Quality: 0.8 (0.6) Duration: 0.7 (1.0) Latency: 1.0 (1.1)	-

First author, Year	Study design	Control group (type, *n* and % female, age(SD) years)	Sleep outcome	Carer burden measure	Sleep Results Mean (SD)	Carer Burden Results Mean (SD)
					Daytime dysfunction: 0.9 (0.6) Sleep efficiency: 0.8 (1.1) GSQ; Excessive daytime sleepiness (17.4% of carers) Excessive dreams (4.3% of carers) Hallucinations (0% of carers)	
Perez, 2020 [44]	Cross-sectional	-	PHQ-9 Sleep questions: Trouble falling to sleep or staying asleep, or sleeping to much" "Feeling tired or having little energy" FNQ “I need to get enough rest or sleep"	ZBI	Not reported	Not reported
Smith, 1997 [48]	Case-control	Healthy controls, *n* = 103, F = 43%, 63.0 (14.0)	Self-report sleep quality: 'I cannot sleep well at night' Disrupted sleep: “I can no longer sleep through the night because my partner needs help (e.g., to turn over, to be covered by a blanket, to go to the toilet)”	-	**Carers;** Sleep quality: 2.3 (1.0) Sleep disturbance Female: 2.3 (1.0) Males: 1.9 (0.9) **Control;** Sleep quality: 1.8 (0.9)	-

First author, Year	Study design	Control group (type, *n* and % female, age(SD) years)	Sleep outcome	Carer burden measure	Sleep Results Mean (SD)	Carer Burden Results Mean (SD)
Teel, 1999 [49]	Case-control	i) Healthy controls, *n* = 33, F = 49%, 74 ii) AZ Carers *n* = 30, F = 53%, 72 iii) Cancer, *n* = 33, F = 52%, 70	Sleep disturbance: VSH Sleep effectiveness: VSH	-	**PD Carers;** Sleep disturbance: 3.9 Sleep effectiveness: 6.7 **AZ Carers;** Sleep disturbance: 4.5 Sleep effectiveness: 6.9 **Cancer Carers;** Sleep disturbance: 3.9 Sleep effectiveness: 6.9 **Control:** Sleep disturbance: 3.0 Sleep effectiveness: 8.0	-
Thommessen, 2002 [45]	Cross-sectional	-	Sleep disturbance: Is your sleep interrupted by your relative?	-	Sleep disturbance: 17% of carers	-
Wade, 2020 [46]	Cross-sectional	-	PSQI	ZBI	Global PSQI: 6.9 (3.9) Disturbance: 1.3 (0.5) Quality: 0.8 (0.6) Duration: 0.8 (0.9) Latency: 1.0 (1.0) Daytime dysfunction: 1.1 (1.0) Sleep efficiency: 1.1 (1.1)	ZBI: 18.5 (14.2)

Notes: PD; Parkinson's Disease, CES-D, Centre for Epidemiologic Studies Depression Scale-10 item; CBI, Caregiver Burden Inventory; MOS-SS index II, Medical Outcomes Study-Sleep Scale Index II; FNQ, Family Needs Questionnaire; PSQI, Pittsburgh Sleep Quality Index; GSQ, General Sleep Questionnaire, PADRECC, Parkinson's disease research, education and clinical centre; ZBI, Zarit burden interview; HADS, Hospital anxiety and depression scale; GQS, General questionnaire for sleep; SII, Sleep intervention index; SCOPA-sleep, Scale for outcomes in Parkinson's disease – sleep; VSH, Verran and Snyder-Halpern sleep scale; RSS, Relative stress scale; WHOQOL-BREF, World health organisation quality of life assessment-brief; NHP – sleep section, Nottingham health profile survey – sleep section; FTLD, Frontotemporal Lobar Degeneration; AZ, Alzheimer's disease; PMDA, Parkinson and Movement Disorder Alliance survey.